
Reports on NGSOTI training experience and data set

NGSOTI Project: 101127921
DIGITAL-ECCC-2022-CYBER-03 D3.3

UNILU/NGSOTI



Co-funded by
the European Union



ECCC 
EUROPEAN CYBERSECURITY
COMPETENCE CENTRE

2026-06-30

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1 NGSOTI training experience and data

1.1 Disclaimer

Co-funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Cybersecurity Competence Centre. Neither the European Union nor the granting authority can be held responsible for them.

1.2 Distribution and License

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The document is distributed as TLP:CLEAR.

1.3 Deliverable Definition

The identifier of the deliverable is D3.3 and it adheres to the definition outlined in the grant agreement **Reports on NGSOTI training experience and data**. The deliverable name is **Reports on NGSOTI training experience and data** and the overall objective/alignment is described in the executive summary.

1.4 Executive Summary

This report evaluates the NGSOTI training delivered to a cohort of graduate-level cybersecurity students and documents the anonymised data collected through the accompanying post-training survey. The cohort comprises seventeen participants ($n = 17$), each attending one or more of seven hands-on sessions on MISP (CTI fundamentals and automation), Lookyloo, SkillAegis, Kunai, the AIL Framework, and digital forensics. Participants complete a structured evaluation questionnaire that combines **retrospective pre/post self-assessments** with reaction-level and behaviour-level items, designed around **Kirkpatrick's four-level training evaluation model** (Reaction, Learning, Behavior, Results); this report follows the same structure to present its findings.

Across the five tools, self-reported familiarity rises by an average of **+1.46 points on a 5-point scale** (from a pre-training mean of 1.52 to a post-training mean of 2.97), and confidence in understanding and applying the tools improves by **+0.89 to +1.41 points** depending on the tool and dimension. Participants rate the training positively on usefulness (mean 4.06/5) and clarity (3.88/5), and a majority report concrete plans to continue working with the tools (independently, academically, or professionally) within the following one to three months. The most frequently cited obstacle to applying what they learn is **insufficient personal knowledge to proceed independently**, which indicates that the appetite for the material extends beyond a single training cycle and supports the case for continued and expanded delivery.

2 Methodology

2.1 Training context

The evaluated training is delivered to a cohort of students of the Master in Cybersecurity and Cyber Defence (MCYSD) at the University of Luxembourg, as part of the academic activities supported under NGSOTI Trainings. The cohort consists of graduate-level students specialising in cybersecurity and cyber defence, a characteristic that shapes both the level of prior knowledge participants bring to the sessions and the kinds of follow-on activities (university projects, theses, internships) that the Behavior-level findings describe. The programme combines instructor-led explanation, live demonstrations, and hands-on exercises across seven tracks:

- **MISP CTI**: fundamentals of the Malware Information Sharing Platform and threat intelligence workflows
- **MISP Automation**: scripting, automation, and integration around MISP
- **CTI at Large**: broader cyber threat intelligence concepts and practice
- **AIL Framework**: information-leak analysis and OSINT pivoting

- **Forensics:** digital forensics tooling, disk imaging, and log/host analysis
- **Kunai:** runtime threat detection and monitoring

All tools used in the training are open-source components of the NGSOTI technical stack (MISP, AIL, Lookyloo, Kunai, SkillAegis), giving participants direct exposure to the same software used operationally by SOC teams and CSIRTs in the consortium and beyond.

2.2 The Kirkpatrick model as the evaluation framework

The post-training questionnaire is designed around **Kirkpatrick's Four-Level Training Evaluation Model**, a widely used framework for assessing the effectiveness of learning programmes. The model examines training outcomes at four progressively deeper levels:

1. **Reaction:** how participants experience the training: whether they find it engaging, clear, and relevant
2. **Learning:** what knowledge, skills, and confidence participants acquire
3. **Behavior:** whether participants intend to, or already, apply what they learned in their studies, projects, or professional context
4. **Results:** the longer-term, organisational-level outcomes attributable to the training (for example, measurable improvements in CTI capability across the wider ecosystem)

The first three levels map directly onto sections of the questionnaire: Reaction items cover satisfaction, clarity, usefulness, difficulty match, and engagement; Learning items cover the pre/post familiarity and confidence batteries and the self-rated increase in CTI understanding; and behavior items cover planned next steps and barriers to application. Level 4 (Results) sits beyond the scope of a single post-event survey as it requires longitudinal observation across cohorts, which is the purpose the collected data described in this report, as it is designed to serve a structured, comparable baseline to track over time as NGSOTI training scales up.

2.3 Instrument and data collection

Data collection takes place through a single, anonymous, self-administered online questionnaire (Microsoft Forms), completed after the training. The questionnaire comprises:

- **Retrospective pre/post items:** for each of five tools (MISP, Lookyloo, SkillAegis, Kunai, AIL), participants rate their *familiarity* and their confidence in understanding and applying the tool both **before** and **after** the training, on 5-point Likert scales. This retrospective design avoids the response-shift bias that affects traditional pre-test/post-test comparisons, where participants tend to overestimate their initial competence before they know what they don't know, and keeps the questionnaire short enough to complete in a single sitting.

- **Reaction items:** 5-point ratings of overall satisfaction, clarity and usefulness of explanations and demonstrations, match between training difficulty and participants' level, motivation to explore further, and open questions on the most engaging element and on what could be improved.
- **Learning items:** the pre/post batteries above, plus a direct 5-point self-rating of how much the training increases participants' understanding of cyber threat intelligence, and open questions on the most valuable tool or activity and on topics that deserve deeper coverage.
- **Behavior items:** a multiple-choice question on concrete actions participants plan to take in the following one to three months (for example, setting up a tool independently, using it in a university project, applying for an internship), and a question on perceived barriers to doing so.
- **Demographic items:** age, gender, highest level of education, nationality, and professional or educational background, collected to characterise the sample rather than to identify individuals.

The questionnaire's sections are themselves labelled after Kirkpatrick's first three levels (Reaction, Learning, Behavior) giving a solid scientific basis for the training experience evaluation. All figures and statistics in this report are drawn from the data collected through the questionnaire.

3 Participant Profile

The cohort consists of 17 participants with reasonably diverse backgrounds. Median age is 26 years (range 23–40). The group skews male, consists predominantly of students at Master's level, and combines EU and non-EU nationalities. Participants' professional and educational backgrounds cluster around computer science/software engineering and cybersecurity/information security, consistent with the target MCYSD audience, with smaller numbers reporting network engineering or data-science backgrounds.

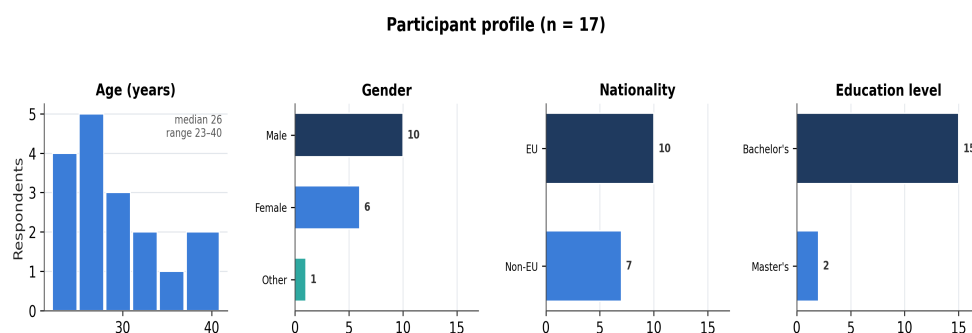


Figure 1: Participant profile: age distribution, gender, nationality, and highest education level (n = 17)

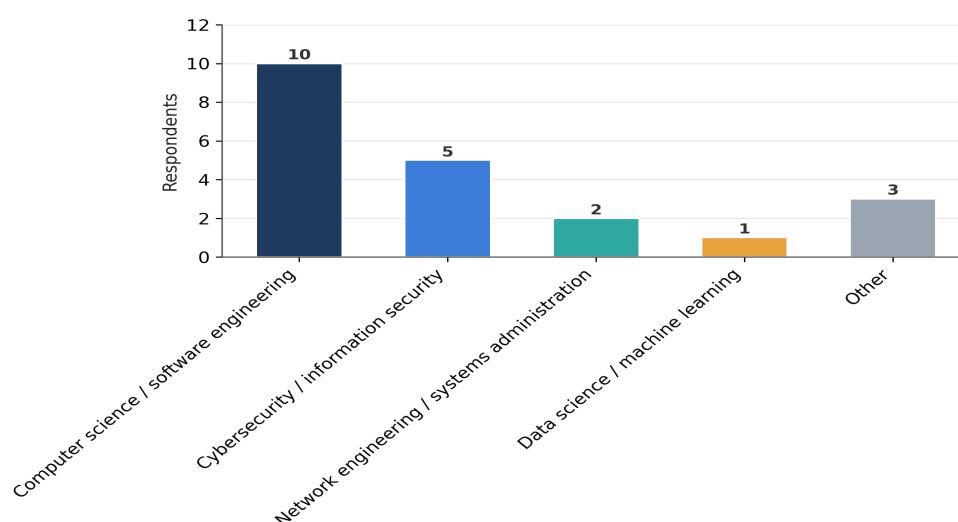


Figure 2: Professional and educational background of participants (multiple selections allowed)

Attendance is high and broadly even across the seven tracks: every session is attended by at least 15 of the 17 respondents, with MISP CTI drawing full attendance (17/17). This even spread means that the pre/post comparisons below reflect the cohort's overall experience of the programme rather than the experience of a narrow sub-group.

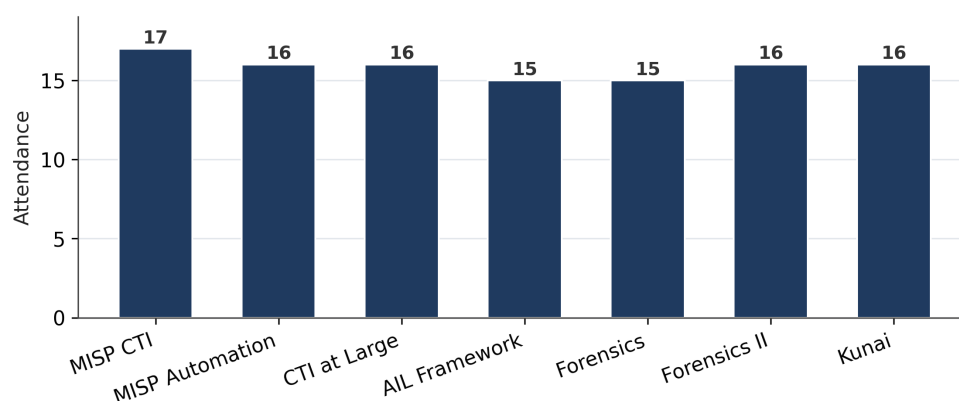


Figure 3: Session attendance across the seven NGSOTI training tracks (n = 17)

4 Reaction (Level 1): How participants experience the training

Reaction-level results are consistently positive. Usefulness of the explanations and demonstrations is the highest-rated dimension (mean 4.06/5), while difficulty match sits closest to the scale midpoint (3.35/5), indicating that the material is pitched neither too easily nor too demandingly for this cohort.

Dimension	Mean (1–5)	Median	Distribution (rating: count)
Usefulness of explanations/demos	4.06	4	2:2, 3:3, 4:4, 5:8
Clarity of explanations/demos	3.88	4	2:2, 3:3, 4:7, 5:5
Overall satisfaction	3.59	4	2:2, 3:6, 4:6, 5:3
Motivation to explore further	3.59	4	2:3, 3:4, 4:7, 5:3
Difficulty match (1=too easy, 5=too difficult)	3.35	3	2:1, 3:10, 4:5, 5:1

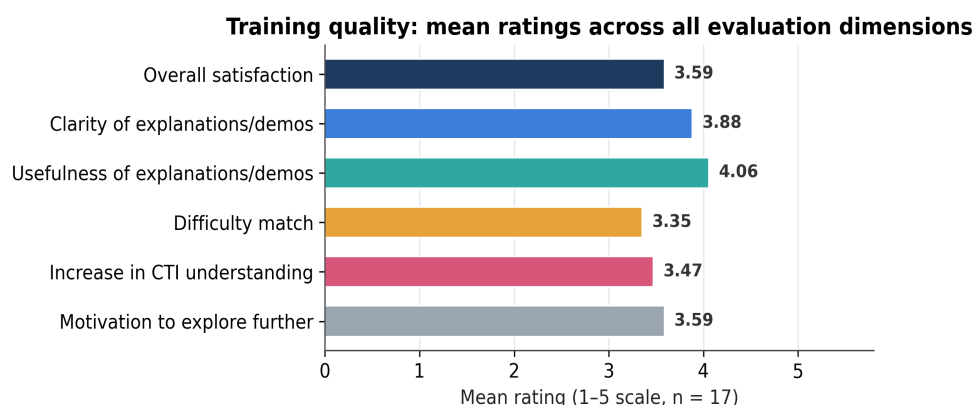


Figure 4: Training quality — mean ratings across all evaluation dimensions (n = 17)

Two-thirds of respondents (12/17) rate overall satisfaction at 4 or 5 out of 5, and the same proportion rates usefulness at 4 or 5. The difficulty-match distribution clusters tightly around “3 – about right” (10 of 17 respondents), indicating that the level of the material is generally well matched to a graduate cybersecurity audience, with a small minority finding it on the easier or harder side.

Asked which element of the training they find most engaging, participants point overwhelmingly to active, applied formats:

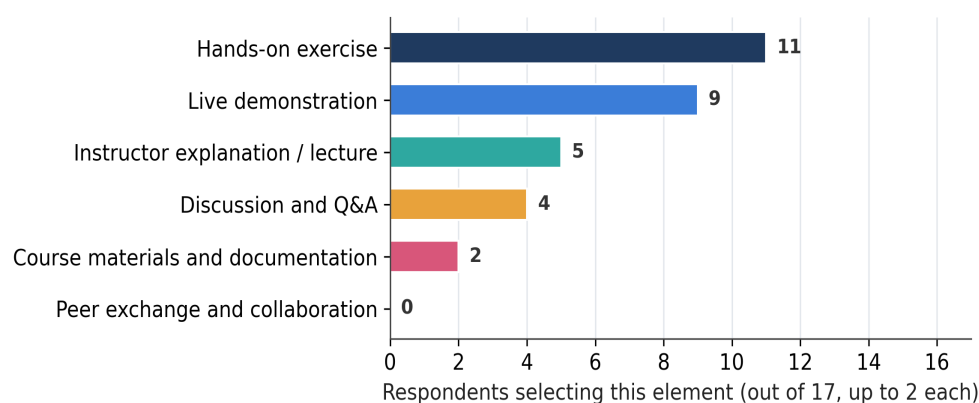


Figure 5: Most engaging training elements, as rated by participants

“Hands-on exercise” (11/17) and “Live demonstration” (9/17) are selected far more often than “Instructor explanation/lecture” (5/17), “Discussion and Q&A” (4/17), or “Course materials and documentation” (2/17). The same pattern recurs in the open-text answers: asked what could be improved, the most frequent suggestion, present in roughly half of all responses, is simply **more hands-on practice and more real-world application**, for example:

“More practical content, how to apply in real situations”

“I would prefer shorter lecture explanations and more hands-on exercises”

“Take away homework to [practice] what we have learned”

5 Learning (Level 2): What participants gain

Across **all five tools and all three dimensions measured** (familiarity, confidence in understanding, confidence in applying), the post-training mean exceeds the pre-training mean, which is a consistent signal of self-perceived learning gain that the retrospective pre/post design is intended to capture.

5.1 Familiarity with the tools

Tool	Pre (mean)	Post (mean)	delta (Δ)
Kunai	1.65	3.18	+1.53
Lookyloo	1.47	2.94	+1.47
SkillAegis	1.06	2.53	+1.47
MISP	1.94	3.35	+1.41
AIL	1.47	2.82	+1.35

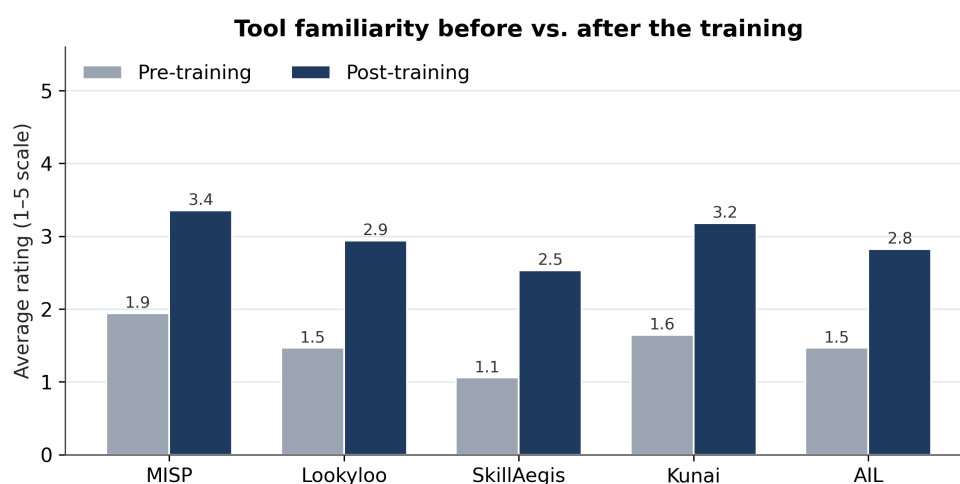


Figure 6: Tool familiarity before vs. after the training (mean rating, n = 17)

The largest relative gain occurs for SkillAegis, which starts from the lowest pre-training baseline (1.06, i.e. participants report being essentially unfamiliar with it) and still records one of the largest absolute increases. MISP reaches the highest post-training familiarity in absolute terms (3.35/5), consistent with its two dedicated sessions (MISP CTI and MISP Automation) and its presence across other tracks such as CTI at Large and AIL.

5.2 Confidence in understanding and applying the tools

Tool	Understanding Δ	Applying Δ
MISP	+1.29	+1.41
Kunai	+1.35	+1.18
AIL	+0.94	+1.12
Lookyloo	+1.12	+1.06
SkillAegis	+0.76	+0.71

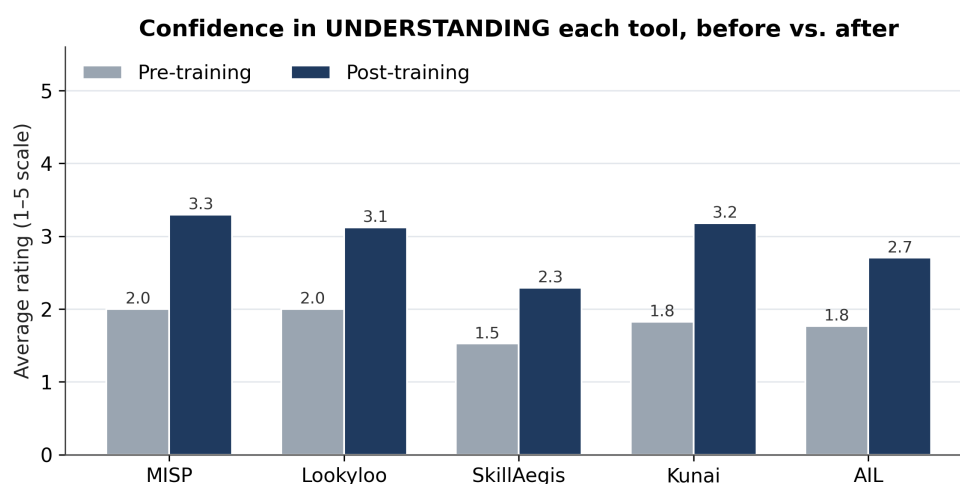


Figure 7: Confidence in UNDERSTANDING each tool, before vs. after (mean rating, n = 17)

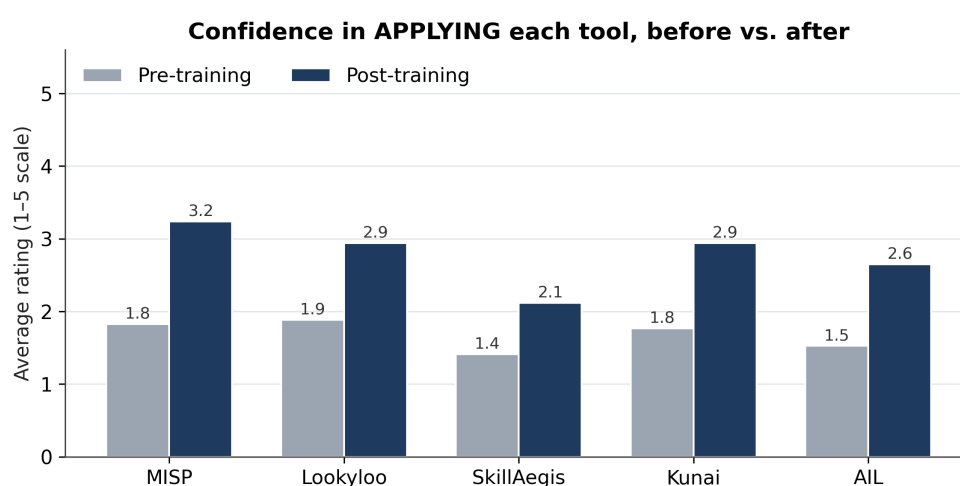


Figure 8: Confidence in APPLYING each tool, before vs. after (mean rating, n = 17)

Participants show increased confidence after the sessions, with MISP and Kunai show the strongest improvements on every dimension, while SkillAegis shows the smallest - though still positive — gains in confidence despite a strong familiarity gain. This is consistent with a tool that participants now recognise but have not yet had enough hands-on time to feel they could operate independently, precisely the gap that additional exercise time (participants' own leading suggestion; see Reaction) would be expected to close. For four of the five tools, gains in “applying” confidence are at least as large as gains in “understanding” confidence, indicating that the training succeeds not only in explaining concepts but in giving participants enough practical exposure to put the tools to work rather than merely describe and present them.

5.3 Self-rated learning increase and open feedback

Asked directly how much the training increases their understanding of cyber threat intelligence, participants report a median of 4/5 and a mean of 3.47/5; 9 of 17 rate the increase at 4 or 5. Open-text answers about the most valuable activity converge heavily on **MISP** (named directly by roughly half of respondents) and on **hands-on, scenario-based exercises**:

“MISP, we had live demonstration and hands-on real-world incident to apply what we had learnt”

“The hands-on during the forensics classes where we created disk images and did forensic analysis on them ... it improved my knowledge”

“Kunai and threat intel” / “kunai, it was completely new and challenging for me so I had to study hard to get into it”

Asked which topic deserves deeper coverage, **CTI** and **MISP** are the two most frequently named, alongside several requests for more coverage of “the code”, “the project architecture”, and end-to-end real-case workflows, which is a concrete pointer for refining future iterations of the curriculum.

6 Behavior (Level 3): What participants intend to do next

Looking forward, participants report concrete intentions to continue engaging with the tools and material covered in the training, alongside the obstacles they expect to encounter.

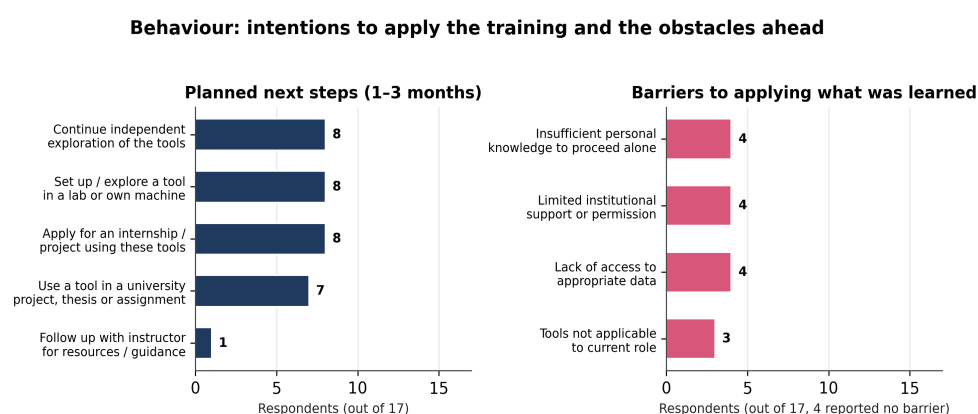


Figure 9: Behaviour: intentions to apply the training and the obstacles ahead

Eight of seventeen respondents (47%) plan to **continue independently exploring the tools**, and the same number plan to **set up and explore a tool in a lab or on their own machine**, in the next one to three months; eight also intend to **apply for an internship or project involving these tools**, and seven plan to **use a tool in a university project, thesis, or assignment**. Most respondents select more than one of these options, indicating a generally high level of post-training engagement rather than a binary interested/not-interested split.

Among barriers, the most frequently cited by a clear margin is “**insufficient personal knowledge to proceed independently**” (5 mentions), followed by “**limited institutional support or permission**” and “**lack of access to appropriate data**” (4 mentions each), and “**tools not applicable to my current role**” (3 mentions). Three respondents report perceiving no barrier at all (“Nothing”, “I don’t think so”, “Nothing like that”).

Reading through the Learning results, the leading barrier “insufficient personal knowledge” is not a contradiction but a natural consequence of a successful introductory training: participants now know enough to recognise what they do not yet know, and to want to close that gap (the second most common open-text request, after more hands-on time, amounts to “more of this”). This combination of rising familiarity, rising confidence, and self-aware appetite for more, is the trajectory a training programme aims to produce on first encounter with a cohort, and it points to continuity (follow-up sessions, lab access, mentoring) as the most effective direction for converting intention into applied behaviour.

7 Discussion

The four levels of the Kirkpatrick model, taken together, describe a coherent pattern for this cohort. Reaction is solidly positive, with the clearest signal being a strong preference for hands-on, applied formats over lecture-style delivery, paired with an equally clear request for more of the same. Learning gains are consistent and broad-based: every tool, on every dimension measured, moves in the expected direction, with the largest gains concentrated on the tools that receive the most session time and hands-on exposure (MISP, Kunai). Behavior intentions are concrete and largely academic or professional in nature (lab exploration, university projects, internships) the kind of follow-through expected from a graduate cybersecurity cohort, and one that feeds directly into the longer-term, ecosystem-level outcomes that **Results (Level 4)** addresses.

As a main recommendation that follow directly from the data, we report the **Increase the proportion of hands-on exercise time**. This is the most consistent request across both the closed-ended “engaging element” question and the open-text improvement suggestions, and it coincides with where the largest remaining confidence gaps sit (e.g. SkillAegis “applying” confidence).

8 data Description

Field groups.

Group	Fields	Notes
Demographics	age, gender, education level, nationality, professional/educational background	Categorical / ordinal; used to characterise, not identify, the sample
Attendance	session(s) attended	Multi-select across 7 tracks
Pre-training self-assessment	familiarity and confidence (understanding / applying), per tool, for 5 tools	5-point Likert, retrospective pre-measure
Post-training self-assessment	familiarity and confidence (understanding / applying), per tool, for 5 tools; self-rated CTI-understanding increase	5-point Likert
Reaction	satisfaction, clarity, usefulness, difficulty match, motivation to explore further, most engaging element	5-point Likert + multiple-choice
Behavior	planned actions in the next 1–3 months; perceived barriers	Multiple-choice, multi-select
Open-text	improvement suggestions (clarity / delivery), highest-value session, most valuable tool/activity, topic for deeper coverage, final feedback	Free text; useful for thematic/qualitative analysis as illustrated in this report

Future reuse. The structure of this data is designed to be **repeatable**: future training cohorts can be surveyed with the same instrument, and their anonymised responses appended to or compared against this baseline, supporting longitudinal tracking of learning-gain trends, and cross-cohort comparisons.

9 Annex: Survey Instrument

The questionnaire fielded for this cohort comprises the following sections and item types (verbatim question wording is preserved in the released data's column headers):

1. **Demographics:** age, gender, highest level of education, nationality, professional/educational background, session(s) attended.
2. **Pre-training self-assessment:** for each of MISP, Lookyloo, SkillAegis, Kunai, and AIL: familiarity (5-point scale), confidence in understanding (5-point scale), confidence in applying (5-point scale).
3. **Reaction:** overall satisfaction, clarity of explanations/demonstrations, usefulness of explanations/ demonstrations, match between difficulty and professional/knowledge level, most engaging training element (select up to 2), open questions on what could be improved (clarity, delivery), and which session had the highest added value.
4. **Post-training self-assessment:** the same familiarity and confidence batteries as the pre-training section, repeated for the same five tools, plus a direct self-rating of how much the training increases understanding of cyber threat intelligence, an open question on the most valuable tool/activity, and an open question on which concept or skill deserves deeper coverage.
5. **Behavior:** motivation to explore the topics/tools further, planned actions in the next one to three months (select all that apply), perceived barriers to applying what was learned (select all that apply), and final open-ended feedback.